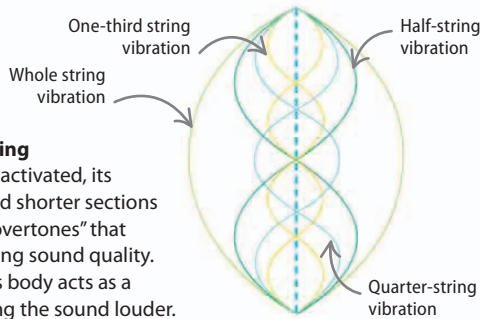


Producing the sound

All stringed instruments produce sound through the vibration of their strings. The strings of the guitar and harp are plucked, while the orchestral strings—violin, viola, cello, and double bass—are vibrated with a bow.



▷ Vibrating string

When a string is activated, its whole length and shorter sections vibrate, giving “overtones” that produce a pleasing sound quality. The instrument’s body acts as a resonator, making the sound louder.

LOOKING CLOSER

String techniques

While the guitar and harp are always played with the fingers, orchestral strings are played with a bow, or “con arco.” Other techniques can be used to vary the sound and produce special effects, ranging from comic to spooky.



◁ Bowed

The direction in which the player draws the bow across a string is indicated by these symbols.



◁ Tremolo

To play tremolo, the player moves the bow in rapid short strokes across the string.



◁ Portamento

This swooping effect is created between two notes when the finger slides up the fingerboard.

pizz.

△ Plucked

When playing “pizzicato,” the player leaves the bow aside and plucks the string.

con sord.

△ Muted

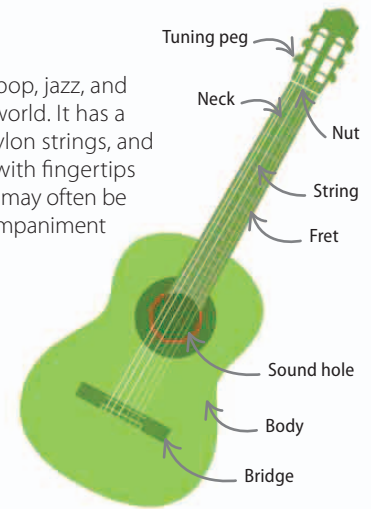
To deaden the sound, a three-pronged wooden mute (sordino) is clamped across the bridge of the instrument.

Guitar

The guitar is heard in folk, pop, jazz, and classical music across the world. It has a violin-type body, wire or nylon strings, and a fretted neck. It is played with fingertips or a plectrum (pick), and it may often be heard as a strumming accompaniment to a song or a dance.

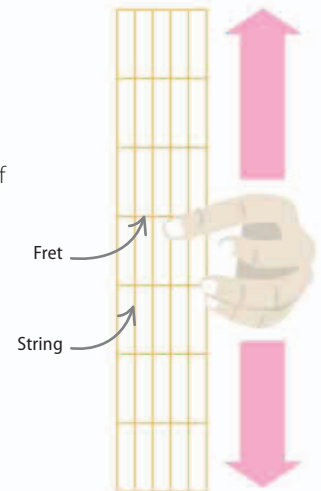
▷ Acoustic guitar

The acoustic guitar has long associations with Spain, especially with the gypsy-inspired flamenco dance music in which the guitar body can be slapped loudly like a drum.



Changing the pitch

The pitch of a string is changed by altering its length, or by increasing or decreasing its tension. If the string is shortened or tightened, the pitch is raised; if it is lengthened or loosened, the pitch is lowered.

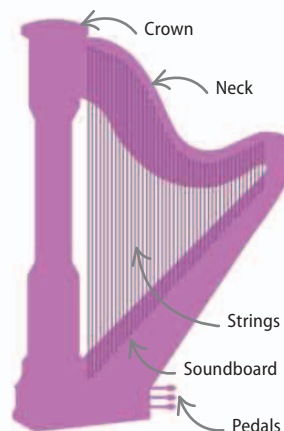


▷ Stopping a string

Pressing down the fingertip on the fingerboard shortens the string and changes its pitch. Guitar necks have narrow raised frets indicating where the player should press.

Harp

Harpes have sturdy frames and a resonating chamber with a soundboard to project the sound. Strings are played with the thumb and fingers of both hands, though the little finger is not used. Small harps used in folk music may rest in the player’s lap.



◁ Concert harp

The large concert harp has 47 strings of different lengths. The pitch of any string can be raised by a half or whole step by pressing down a foot pedal.